CNN

- $*$ convolution $\quad y[n] = h[n] * x[n]$

$$= \sum_{k=0}^{M} h[n] \times x[n-k]$$

- correlation $\quad g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\, f(x+s, y+t)$

- convolution $\quad g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\, f(x-s, y-t)$

↳ commutative (with flip)

- convolution 2D (cross correlation)

$$S(i,j) = (I * K)(i,j) = \sum_m \sum_n I(m,n)\, K(i-m, j-n)$$

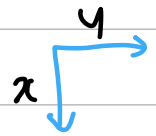
$$\| \quad \text{(commutative)}$$

$$(K * I)(i,j) = \sum_m \sum_n I(i-m, j-n)\, K(m,n)$$

↓ correlation

$$S(i,j) = (K * I)(i,j) = \sum_m \sum_n K(m,n)\, I(i+m, j+m)$$

- Sparse connectivity, receptive field

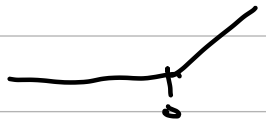


- zero padding

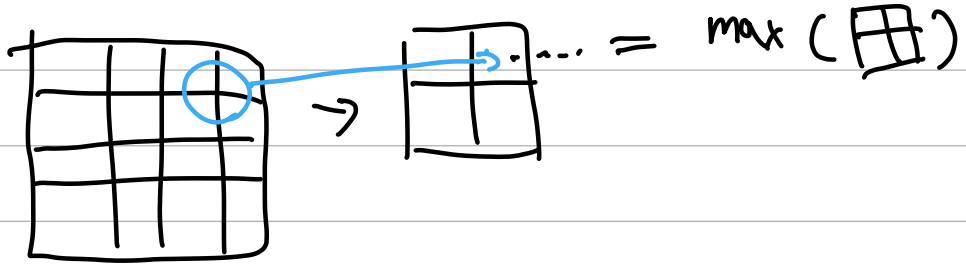
Padding 없으면 계속 -1 만큼 줄음.

- rectified Linear Unit

$$f(z) = \begin{cases} z, & z \geq 0 \\ 0 & z < 0 \end{cases}$$



- pooling

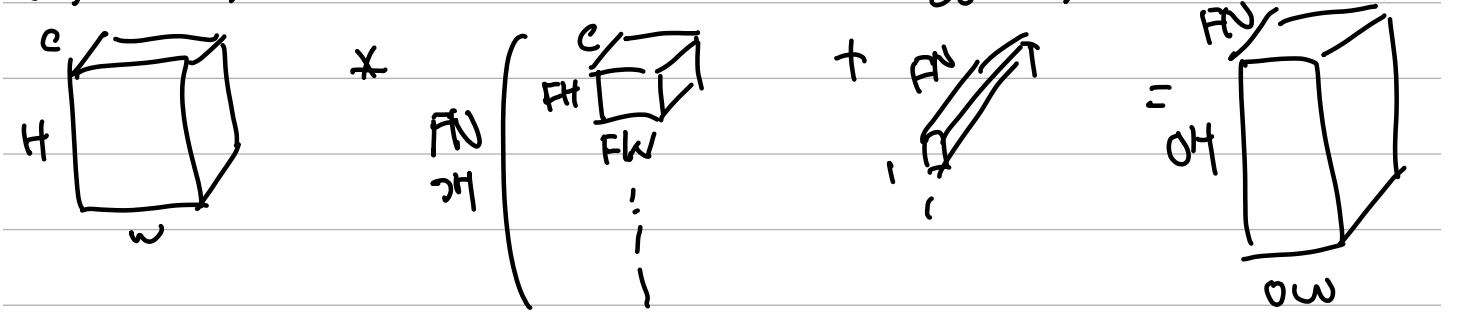


- Stride

이동하는 간격.

- 3D

입력 (C, H, W) 필터 (FN, C, FH, FW) 편향 (FN, 1, 1) 결과 (FN OH, OW)



$$\sum_{l}\sum_{m}\sum_{n} V_{l,j+m,k+n} K_{j,l,m,n} = z_{i,j,k}$$

$$z_{ijk} = \sum_{d} \sum_{m} \sum_{n} V_{d,j+m,k+n} \; K_{i,d,m,n}$$